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2.2.6 Change of orthonormal basis

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unitary matrix, we know that that's the same as u times u Hermitian transpose times the vector x . Okay? Oh, well, we're starting to get close because this u here is what we have right here, and U Hermitian transpose times x is a vector, which then should give us the coefficients that we desire. And now we can go into a little bit more detail and say, "Well that's just equal to u times the matrix U written as its columns but transposed so that gives us u_0 Hermitian transpose as

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Reading assignment

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Read Unit 2.2.6 of the notes. [\[LINK\]](#)

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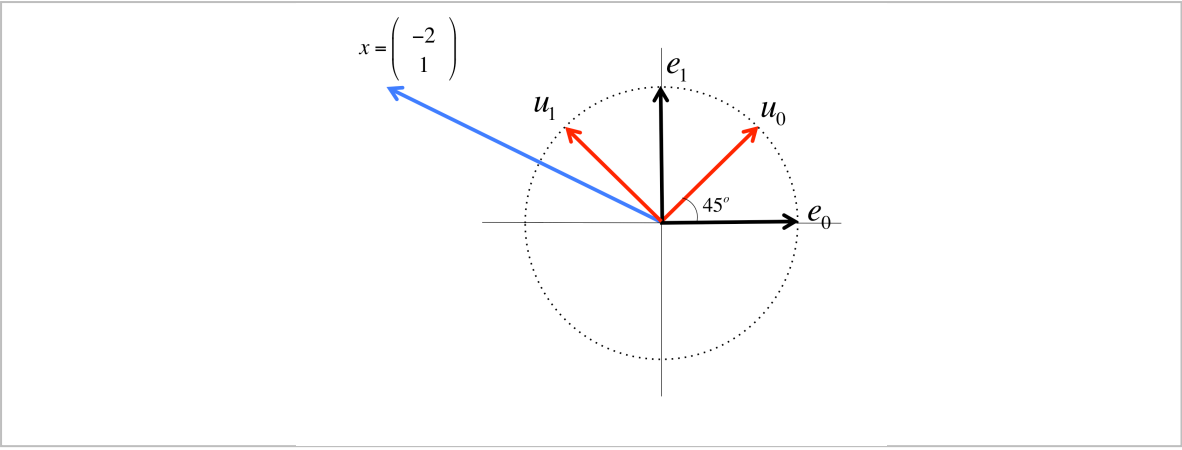
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Homework 2.2.6.1

1/1 point (graded)

Consider the vector $\boldsymbol{x} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$ and the following picture that depicts a rotated basis with basis vectors $\boldsymbol{u}_0$ and $\boldsymbol{u}_1$.



What are the coordinates of the vector $\boldsymbol{x}$ in this rotated system? In other words, find $\hat{\boldsymbol{x}} = \begin{pmatrix} \hat{x}_0 \\ \hat{x}_1 \end{pmatrix}$ such that $\hat{x}_0 \boldsymbol{u}_0 + \hat{x}_1 \boldsymbol{u}_1 = \boldsymbol{x}$.

$\hat{x}_0 =$ ✔ Answer: -sqrt(1/2)

$\hat{x}_1 =$ ✔ Answer: 3*sqrt(1/2)

Solution:

For complete solution, see [[LINK](#)].

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i Answers are displayed within the problem

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